

Loss Aversion, Endogenous Reference Points, and Boom-Bust Asymmetry in Financial Wealth Dynamics

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PhD Economics, University of Reading (2024)

April 2026 — Working Paper

Abstract

Financial markets exhibit a well-documented asymmetry between expansions and contractions: boom periods are volatile, turbulent, and fast; bust periods are compressed, slow, and grinding. This pattern is robust across asset classes, time periods, and economies, yet lacks a structural explanation rooted in individual preferences. This paper derives the asymmetry from first principles using the minimal behavioural structure sufficient to deliver it as an equilibrium theorem rather than a calibration choice.

We retain the single empirically robust element of prospect theory — the kink in the value function at a reference point — and discard the rest. The reference point is the economy-wide Schumpeterian wealth trend, publicly observed and growing at a constant rate. These ingredients are embedded in a Stackelberg market with a loss-averse market maker and a rational speculator, scoped to liquid exchange-traded settings where adverse selection is attenuated and dealer risk management is the primary source of price impact asymmetry.

Two structural features are jointly necessary. First, the zero-sum trading game guarantees as a theorem that the two agents always sit on opposite sides of the kink simultaneously, generating asymmetric price impact every period. Second, the Schumpeterian trend provides mean-reversion that pins volatility persistence to the economy's long-run growth rate. The paper derives a structural recursion for the conditional variance of detrended wealth deviations, with persistence determined by the growth rate alone and the asymmetry ratio by dealer loss aversion alone — a separation property that is the paper's primary identification contribution. All structural mappings are derived analytically and verified symbolically.

Keywords: loss aversion; boom-bust asymmetry; Threshold-GARCH; market microstructure; Stackelberg equilibrium; Schumpeterian growth

1 Motivation

1.1 The empirical pattern this paper explains

One of the most robust patterns in financial history is the asymmetry between boom and bust in the dynamics of financial wealth. Expansions — periods when aggregate dealer or investor wealth sits above its long-run growth trend — are characterised by rapid balance sheet growth, large position sizes, elevated volatility, and turbulent price movements. Contractions — periods of below-trend wealth — are slow, compressed, and grinding.

Kindleberger (1978) documents this pattern across two centuries of financial manias and crashes: the run-up to a crisis is invariably turbulent and fast, the aftermath slow and drawn out. Adrian and Shin (2010) show that financial intermediary balance sheets expand rapidly and with high variance when dealers are above their funding benchmarks, and contract slowly when approaching their limits. Brunnermeier and Pedersen (2009) establish that funding liquidity constraints generate precisely this asymmetry at the position level: above the constraint, dealer positions are large and wealth dynamics are turbulent; at the constraint, positions compress and wealth process volatility falls sharply. Reinhart and Rogoff (2009), covering 66 countries across eight centuries, document the same pattern at the macro level — the run-up to a financial crisis is associated with high and rising wealth volatility, while the aftermath is characterised by persistent below-trend deviations with compressed variance. Danielsson and Zigrand (2008) show theoretically and empirically that endogenous risk — generated by VaR-constrained intermediaries responding to their own wealth positions — is higher in expansions than contractions, consistent with a structural variance asymmetry that has its source in dealer behaviour rather than in the distribution of fundamentals.

The pattern is well described but structurally unexplained. Existing models either take it as given — describing it statistically without deriving it from preferences — or generate it from mechanisms that require strong distributional assumptions about probability perception incompatible with rational market behaviour. The present paper identifies the minimal structure that delivers the asymmetry and its persistence as equilibrium theorems.

1.2 Why Cumulative Prospect Theory is over-specified for this purpose

The natural behavioural candidate for a structural explanation is Cumulative Prospect Theory (Tversky and Kahneman (1992)). CPT generates asymmetric responses to gains and losses through two mechanisms: the kink in the value function at the reference point, which makes losses loom larger than equivalent gains, and the probability weighting function, which distorts small probabilities upward and large probabilities downward. In

static settings with exogenous reference points, CPT is empirically well supported. In a dynamic financial market equilibrium, however, it is analytically over-specified.

The probability weighting function is rank-dependent and non-differentiable, making it incompatible with recursive dynamic modelling. Dynamic consistency fails: a CPT agent's preferences over plans change as the plan is executed, creating time-inconsistency that cannot be resolved without additional assumptions. Aggregation across CPT agents with heterogeneous reference points and probability weighting functions generates indeterminate equilibria. Most damagingly for the present purpose, the reference point is either exogenous — which is indefensible in a market where agents form beliefs and positions endogenously — or endogenised through a rational expectations fixed-point recursion (Kőszegi and Rabin (2006, 2007), 2006, 2007) that becomes intractable when the reference point and equilibrium wealth volatility are jointly determined.

Ang et al. (2005) and Routledge and Zin (2010) show that many of CPT's empirical successes can be replicated by retaining only the kink — through disappointment aversion or related specifications — while discarding the probability weighting function. This paper takes the same approach but goes further: it also replaces the rational expectations reference point with a simpler social benchmark, eliminating the fixed-point problem at its source.

The most closely related paper in the behavioural asset pricing literature is Barberis et al. (2001), who place loss aversion over financial wealth fluctuations at the centre of a representative agent consumption-based asset pricing model and generate time-varying risk aversion through the house money effect — after prior gains the investor is less loss averse, after prior losses more so. Four differences distinguish this paper from that contribution. First, loss aversion here is fixed rather than time-varying: the market maker's λ does not change with prior performance. Second, the reference point is a shared social Schumpeterian benchmark rather than an individual historical benchmark that tracks personal investment outcomes. Third, the model is a strategic market microstructure equilibrium with three distinct agents rather than a representative agent consumption CAPM. Fourth, the paper derives exact closed-form structural mappings verified symbolically rather than characterising equilibrium properties through numerical simulation. Barberis et al. (2001) establish that loss aversion over financial wealth can explain the equity premium and excess return volatility at the aggregate level; this paper establishes what fixed dealer loss aversion against a social benchmark delivers in a market microstructure equilibrium.

1.3 The minimal structure

This paper focuses on a specific and empirically important class of markets: liquid, exchange-traded settings where prices are publicly observable, adverse selection is at-

tenuated, and the dominant source of price impact asymmetry is dealer risk management behaviour rather than private information. In these markets, VaR constraints and balance sheet benchmarks govern dealer behaviour more than informational disadvantage, and loss aversion is the structurally appropriate primary mechanism.

We retain the single empirically robust element of prospect theory: the kink in the value function at a reference point, which makes losses loom larger than equivalent gains. Everything else — the probability weighting function, the rank-dependence, the S-shaped curvature away from the kink — is discarded. The reference point is the economy-wide Schumpeterian wealth trend, growing at a constant rate $g > 0$ and publicly observed by all agents. This is the simplest social benchmark consistent with a market setting. It requires no individual wealth histories, no rational expectations fixed-point, and no peer-group comparison. Its merit is that it is the weakest assumption capable of delivering the structural results; its limitation is that it abstracts from heterogeneous benchmarks and time-varying growth.

The theoretical motivation for a social reference point connects to the social comparison literature (Festinger, 1954; Fehr and Schmidt, 1999) and to catching-up-with-the-Joneses models in asset pricing (Abel, 1990; Campbell and Cochrane, 1999). The Schumpeterian implementation is distinct from these contributions: the reference point is endogenous to the system (determined by the economy’s growth trajectory) but exogenous to each individual agent’s decision, eliminating the strategic manipulation of reference points that complicates Kőszegi and Rabin (2006, 2007) implementations.

1.4 Two necessary structural conditions

The structural results depend on two features of the minimal setup that are jointly necessary: each is required for a specific structural result.

The first is the combination of the shared reference point with the zero-sum trading structure and within-period position closing. Together, these three conditions guarantee as a theorem — Proposition 1 below — that the market maker and speculator always sit on opposite sides of the kink simultaneously. When total detrended wealth is conserved, one agent’s gain above the benchmark is exactly the other’s loss below it. The loss-aversion mechanism therefore operates asymmetrically through the price-setter and the quantity-setter in every period, generating the boom-bust variance asymmetry without additional assumptions. If the shared reference point is replaced by own-history benchmarks, this regime separation is lost and the structural link between wealth volatility persistence and the growth rate disappears entirely.

The second is the Schumpeterian growth trend itself, which provides the mean-reversion in the deviation recursion. Each period, detrended wealth reverts toward the benchmark at rate $1/(1 + g)$. This deflation factor is the structural source of persistence in wealth

volatility dynamics: it connects the rate at which volatility shocks dissipate to the economy’s long-run growth rate in a precise and testable way. If the S-shaped value function is removed — that is, if the kink disappears at $\lambda = 1$ — the asymmetry between boom and bust variance vanishes exactly, confirming that loss aversion is not only sufficient but necessary for the result.

The institutional counterpart of the model is a dealer trading desk operating under Value-at-Risk constraints against an informed fund, both benchmarked against the market’s trend growth. Li et al. (2025), using granular desk-level regulatory data on US Treasury dealers, document precisely this mechanism: dealers reduce position size as they approach their internal VaR limits, and accept lower prices to shed inventory when the limit binds. This behaviour is isomorphic to the model’s contraction regime, in which the market maker — above trend and protective of accumulated gains — posts high price impact and the speculator takes small positions.

1.5 Contribution and relation to existing work

The paper makes four contributions. The first is the derivation of boom-bust wealth volatility asymmetry as an equilibrium theorem from the minimal CPT-like structure. The second is the structural identification result: the persistence of wealth volatility is determined by the growth rate alone, and the asymmetry ratio is determined by loss aversion alone, with the two parameters orthogonal. This separation means that two observable features of wealth volatility dynamics separately identify two economic primitives without auxiliary assumptions. The third is the IGARCH boundary result: as the growth rate approaches zero, volatility persistence approaches one, providing the first preference-based microfoundation for integrated volatility processes. The fourth is the connection of the structural mappings to a testable cross-sectional prediction: in cross-country data, persistence estimates should co-vary with long-run trend growth and asymmetry ratios with measures of dealer loss aversion, with zero cross-effects.

The paper is postdictive in intent: it explains why the boom-bust asymmetry in financial wealth takes the specific parametric form it does, given the structural primitives of dealer loss aversion and trend growth, rather than fitting or forecasting it. The natural empirical counterpart — estimating the structural recursion on detrended wealth or dealer balance sheet data and testing whether the cross-sectional restrictions hold — is a direction for future work that the structural results are designed to motivate.

A fifth contribution concerns the relationship between loss aversion and adverse selection. In markets where both channels operate simultaneously, they are observationally entangled in the price impact coefficient: any asymmetry in observed price impact could in principle reflect either the market maker’s informational disadvantage (adverse selection) or its behavioural response to its own wealth position (loss aversion). By shutting down

adverse selection deliberately, this paper isolates the loss aversion channel and establishes that it is sufficient for boom-bust asymmetry even in the absence of private information. The separation property — $\partial\beta/\partial\lambda = 0$ and $\partial(|\gamma|/\alpha)/\partial g = 0$ — then provides the identification strategy for a fuller model in which both channels operate simultaneously: growth policy shifts β without affecting the asymmetry ratio, while macroprudential policy shifts $|\gamma|/\alpha$ without affecting persistence. The two channels are non-redundant and separately identified.

Existing micro-foundations for volatility clustering (Brock and Hommes, 1998; Föllmer et al., 2005; Watanabe, 2008) derive clustering from heterogeneous agents or information structure rather than from individual preferences in an explicit market equilibrium. The present paper is closest in spirit to Watanabe (2008), who embeds stochastic volatility in a Kyle-style microstructure, but the volatility process there enters exogenously through the information structure rather than being derived from preferences.

Remark 1.1 (Market quality across regimes). *Kyle (1985) characterises market quality through three dimensions: depth (the inverse of price impact), tightness (the bid-ask spread), and resilience (the speed of recovery after a shock). Kyle’s model delivers a single static value for each, determined by the signal-to-noise ratio σ_μ/σ_u . This paper extends all three into a regime-switching setting driven by dealer loss aversion and the Schumpeterian growth rate.*

Depth and tightness are regime-dependent. In expansion the effective price impact is λ_p^0/λ — markets are deeper and tighter than the Kyle benchmark by a factor of λ . In contraction it is $\lambda \cdot \lambda_p^0$ — shallower and wider by the same factor. The ratio of expansion depth to contraction depth is λ^2 , providing a structural theory of liquidity procyclicality consistent with Adrian and Shin (2010) and Brunnermeier and Pedersen (2009).

Resilience is where the model adds most relative to Kyle, which does not address it at all. The persistence parameter $\beta = 1/(1 + g)^2$ is a structural measure of recovery speed: high-growth economies have lower β and faster recovery; stagnant economies approach $\beta = 1$, meaning near-permanent impairment of market quality. This connects liquidity resilience directly to macroeconomic growth — a prediction invisible from reduced-form microstructure estimation.

2 The Framework

2.1 Players and roles

Three agents interact in a competitive financial market at each date t :

- **Market maker (MM).** Posts the price impact schedule $\lambda_p > 0$ before trade. Absorbs total order flow. Has a loss-averse value function with kink parameter

$\lambda > 1$ evaluated against the shared Schumpeterian benchmark. Acts as Stackelberg leader.

- **Speculator (S).** Receives private signal $\mu_t \sim \mathcal{N}(0, \sigma_\mu^2)$ about the asset's fundamental value and chooses position size $m_t \in \mathbb{R}$ after observing the posted price impact schedule. Loss-neutral — standard expected profit maximiser. Acts as Stackelberg follower.
- **Noise traders.** The noise trader assumption follows Kyle (1985) and is standard throughout the market microstructure literature: without uninformed order flow, a competitive market-making equilibrium cannot exist because any order would perfectly reveal the speculator's signal and the market maker would always lose to the informed party. Noise traders submit random order flow $u_t \sim \mathcal{N}(0, \sigma_u^2)$, independent of μ_t and ε_t . The stationarity condition derived in Section 9 requires the signal-to-noise ratio σ_μ/σ_u to exceed a threshold that depends on λ and g .

Remark 2.1 (Uncertainty structure). *The uncertainty structure of the model is de Finetti subjective in character. The private signal μ_t represents the speculator's epistemic advantage — knowledge the market maker does not possess — rather than aleatory randomness intrinsic to the world. The noise trader flow u_t is closer to aleatory uncertainty, representing the irreducible aggregation of many independent liquidity-motivated trades whose individual motivations are unobservable to all agents.*

Crucially, the mechanism generating the boom-bust variance asymmetry is not uncertainty-driven. Both agents observe the Schumpeterian trend and the detrended wealth deviation D_{t-1} : the regime is public information. The market maker's asymmetric price impact response is a certain reaction to a publicly known position, not a probabilistic response to private information. This distinguishes the paper's mechanism sharply from Kyle (1985), where asymmetric price impact arises from the market maker's epistemic uncertainty about informed order flow. Here, loss aversion operates on publicly observable states — consistent with the predominance of public price observation in exchange-traded markets where the information asymmetry problem is attenuated and the risk management problem is primary.

Remark 2.2 (Why loss aversion must reside in the market maker). *A loss-averse speculator generates no equilibrium asymmetry. Under a piecewise linear value function, the loss aversion parameter λ appears as a positive scalar multiplying the speculator's objective and cancels exactly in the first-order condition: the optimal position $m_t^* = \mu_t/(2\lambda_p)$ is independent of λ . For loss aversion to affect equilibrium outcomes it must enter through a binding constraint rather than an interior optimum.*

In the Stackelberg game, the market maker's loss aversion manifests as self-imposed regime-dependent bounds on the price impact schedule it posts. In the gain region (con-

traction regime, $D_{t-1} < 0$, MM above the benchmark) the MM imposes an upper cap on price impact, protective of accumulated wealth; in the loss region (expansion regime, $D_{t-1} \geq 0$, MM below the benchmark) it accepts a lower floor, willing to narrow margins to attract order flow and recover toward the benchmark. Both bounds are derived from the MM's constrained optimisation and both bind in equilibrium. Because the speculator's optimal position is the signal divided by twice the posted price impact, any change in the price impact schedule propagates directly and mechanically to the speculator's position size. The leader's binding constraint propagates to the follower's optimal choice; the follower's constraint does not propagate to the leader. This is why asymmetry requires loss aversion in the price-setter.

This assignment is independently grounded in the empirical microstructure literature. Li et al. (2025) document, using granular desk-level regulatory data on US Treasury dealers, that dealers reduce position changes and require higher compensation as they approach their internal VaR limits — precisely the high-price-impact, small-position behaviour of the contraction regime. Adrian and Shin (2010, 2014) document that dealer balance sheet dynamics are procyclical and VaR-constrained at the aggregate level.

Remark 2.3 (Stackelberg timing). *The Stackelberg ordering — market maker as leader, speculator as follower — reflects the institutional reality that dealers post firm quotes before customer order flow arrives, and those quotes constitute binding obligations at the posted price for at least one trading increment (Glosten, 1994; Parlour and Seppi, 2008). The structural necessity of this assignment is established in Remark 2.2: assigning loss aversion to the follower produces no asymmetry, so the Stackelberg ordering is not merely institutionally motivated but structurally required.*

2.2 The shared reference point

The economy grows at a constant Schumpeterian rate $g > 0$. The shared reference point is:

$$\bar{r}_t = \bar{W}_0 \cdot (1 + g)^t \tag{1}$$

where $\bar{W}_0 = (W_0^S + W_0^{MM})/2$ is the initial average wealth of the two agents. This construction is publicly observed, continuously evolving independently of individual positions, and exogenous to each agent's decision while endogenous to the system. It is deliberately simple: no individual wealth histories, no rational expectations fixed-point, and no peer-group comparison.

The Schumpeterian growth trend serves two distinct roles. Its level determines the threshold at which the regime switches — the point at which the MM moves from below-trend (expansion, loss region) to above-trend (contraction, gain region). Its rate g enters the deviation recursion as the mean-reversion speed, pinning volatility persistence to the economy's long-run growth trajectory.

2.3 Detrended wealth and the state variable

Define detrended wealth for each agent by deflating nominal wealth by the trend:

$$\tilde{W}_t^S = \frac{W_t^S}{(1+g)^t}, \quad \tilde{W}_t^{MM} = \frac{W_t^{MM}}{(1+g)^t} \quad (2)$$

The state variable is the speculator's deviation from the shared benchmark:

$$D_t \equiv \tilde{W}_t^S - \bar{W}_0 \quad (3)$$

The sign of D_{t-1} determines the regime entering period t . When $D_{t-1} \geq 0$ the speculator is above the benchmark (expansion regime) and the MM is below it. When $D_{t-1} < 0$ the speculator is below the benchmark (contraction regime) and the MM is above it. The zero-sum condition — that total detrended wealth $\tilde{W}_t^S + \tilde{W}_t^{MM} = 2\bar{W}_0$ is conserved at each date — ensures that the two agents are always in opposite regimes simultaneously. This is established formally in Proposition 4.1 and is the structural source of the variance asymmetry.

3 Price Process and Wealth Dynamics

Total order flow is $y_t = m_t + u_t$. Prices follow:

$$p_t = p_{t-1} + \lambda_p y_t + \varepsilon_t \quad (4)$$

where $\lambda_p > 0$ and $\mathbb{E}_{t-1}[\varepsilon_t] = 0$, $\mathbb{E}_{t-1}[\varepsilon_t^2] = \sigma_\varepsilon^2$. The Kyle (1985) zero-profit benchmark pins:

$$\lambda_p^0 = \frac{\sigma_\mu}{2\sigma_u} \quad (5)$$

The speculator's wealth under position closing is $W_t^S = W_{t-1}^S + \lambda_p m_t^2 + m_t \varepsilon_t$. The zero-sum condition $W_t^S + W_t^{MM} = 2\bar{W}_0(1+g)^t$ gives $D_t + \tilde{\Delta}_t^{MM} = 0$ for all t .

4 Reference Point, Regimes, and Regime Separation

Under the zero-sum condition, the two agents are always on opposite sides of the reference point:

$$\text{Regime (+): } D_t \geq 0 \Leftrightarrow \tilde{\Delta}_t^{MM} \leq 0 \quad (\text{expansion}) \quad (6)$$

$$\text{Regime (-): } D_t < 0 \Leftrightarrow \tilde{\Delta}_t^{MM} > 0 \quad (\text{contraction}) \quad (7)$$

Proposition 4.1 (Regime Separation). *Under zero-sum dynamics, position closing, and*

a shared reference point, the two agents are never simultaneously in the loss region.

Proof. Under zero-sum dynamics $\tilde{\Delta}_t^S + \tilde{\Delta}_t^{MM} = 0$. Simultaneous loss requires both deviations negative, implying their sum is negative — a contradiction. $\square$

5 Martingale Properties and Asymmetric Drift

Proposition 5.1 (Price Martingale). *Under position closing and $\mathbb{E}_{t-1}[\varepsilon_t] = 0$, the price process $\{p_t\}$ is a martingale.*

Proof. From $p_t = p_{t-1} + \lambda_p m_t + \varepsilon_t$ and the MM's competitive zero-profit condition, $\mathbb{E}_{t-1}[p_t] = p_{t-1}$. $\square$

Proposition 5.2 (Asymmetric Drift). *Define $D_+^* = \lambda \sigma_\mu^2(1+g)/(4g\lambda_p^0)$ and $D_-^* = \sigma_\mu^2(1+g)/(4g\lambda\lambda_p^0)$, with $D_+^*/D_-^* = \lambda^2$. Then: (i) in contraction, $\{D_t\}$ is a submartingale; (ii) in expansion, $\{D_t\}$ is a submartingale for $D_{t-1} < D_+^*$ and supermartingale for $D_{t-1} > D_+^*$; (iii) $D_+^* > D_-^* > 0$: the economy has a positive bias; (iv) the deflation factor $1/(1+g)$ is identical across regimes.*

Proof. See Appendix, Proposition A.5. $\square$

6 Counter-Cyclical Variance and the Boom-Bust Mechanism

The core mechanism is direct. In **expansion** ($D_{t-1} \geq 0$): the MM is in the loss region and posts low price impact $\lambda_p^+ = \lambda_p^0/\lambda$; the speculator takes large positions $m^{*+} = \lambda\mu_t/(2\lambda_p^0)$; the variance of D_t is proportional to $(B^+)^2 = \lambda^2/(4\lambda_p^{02})$.

In **contraction** ($D_{t-1} < 0$): the MM posts high price impact $\lambda_p^- = \lambda\lambda_p^0$; the speculator takes small positions $m^{*-} = \mu_t/(2\lambda\lambda_p^0)$; the variance of D_t is proportional to $(B^-)^2 = 1/(4\lambda^2\lambda_p^{02})$.

The variance ratio is $(B^+)^2/(B^-)^2 = \lambda^4$: expansions are λ^4 -times more volatile in D_t than contractions. For $\lambda = 2.25$ (Tversky-Kahneman), this is a factor of 25.6.

This matches the boom-bust asymmetry documented by Kindleberger (1978) and Hamilton (1989): expansions are turbulent and fast, contractions are compressed and grinding. Loss aversion is necessary and sufficient for this asymmetry.

7 Deviation Dynamics

From the wealth recursion and detrending:

$$D_t = \frac{D_{t-1}}{1+g} + A^r \sigma_\mu^2 + B^r \mu_t \varepsilon_t \quad (8)$$

where in expansion $A^+ = \lambda/(4\lambda_p^0)$, $B^+ = \lambda/(2\lambda_p^0)$, and in contraction $A^- = 1/(4\lambda\lambda_p^0)$, $B^- = 1/(2\lambda\lambda_p^0)$. The ratio $A^+/A^- = B^+/B^- = \lambda^2$.

Three channels operate: (i) Schumpeterian deflation $D_{t-1}/(1+g)$ — the structural source of GARCH persistence β ; (ii) positive drift $A^r\sigma_\mu^2$ — higher in expansion by factor λ^2 ; (iii) stochastic component $B^r\mu_t\varepsilon_t$ — higher variance in expansion by factor λ^4 .

8 Equilibrium Derivation

8.1 The MM's Problem and Regime-Dependent Price Impact

The MM maximises $\pi^{MM}(\lambda_p) = \lambda_p\sigma_u^2 - \sigma_\mu^2/(4\lambda_p)$ subject to regime-dependent risk limits. The unconstrained optimum coincides with the Kyle zero-profit benchmark $\lambda_p^0 = \sigma_\mu/(2\sigma_u)$.

Loss aversion modifies the equilibrium through self-imposed binding constraints: in the gain region (contraction), the MM imposes an upper limit $\lambda_p \leq \lambda\lambda_p^0$; in the loss region (expansion), the MM imposes a floor $\lambda_p \geq \lambda_p^0/\lambda$. Both constraints bind, giving the regime-dependent price impact in equation (9).

8.2 Speculator's Optimal Position

The loss-neutral speculator solves $\max_{m_t} m_t\mu_t - \lambda_p^{\text{eff}}m_t^2$, giving:

$$m_t^* = \frac{\mu_t}{2\lambda_p^{\text{eff}}} = \begin{cases} \lambda\mu_t/(2\lambda_p^0) & D_{t-1} \geq 0 \\ \mu_t/(2\lambda\lambda_p^0) & D_{t-1} < 0 \end{cases} \quad (9)$$

Position ratio: $m^{*+}/m^{*-} = \lambda^2$. The speculator's FOC is identical in both regimes; all asymmetry flows from the MM's pricing behaviour.

8.3 The Cancellation Result

The return innovation is $\varepsilon_t^r = \lambda_p^{\text{eff}}m_t^* + \varepsilon_t$. In each regime:

$$\begin{aligned} \text{Expansion:} \quad & \frac{\lambda_p^0}{\lambda} \cdot \frac{\lambda\mu_t}{2\lambda_p^0} + \varepsilon_t = \frac{\mu_t}{2} + \varepsilon_t \\ \text{Contraction:} \quad & \lambda\lambda_p^0 \cdot \frac{\mu_t}{2\lambda\lambda_p^0} + \varepsilon_t = \frac{\mu_t}{2} + \varepsilon_t \end{aligned}$$

The signal component is $\mu_t/2$ in *both* regimes. Therefore $\text{Var}(\varepsilon_t^r \mid \text{regime}) = \sigma_\mu^2/4 + \sigma_\varepsilon^2 \equiv \bar{\sigma}^2$ identically. The one-period conditional variance of returns is regime-invariant. Variance structure in D_t emerges from the regime-dependent variance of the state variable $(B^r)^2$, not from the conditional return variance.

Remark 8.1 (No adverse selection baseline). *The baseline model shuts down adverse selection deliberately. The return innovation $\varepsilon_t^r = \mu_t/2 + \varepsilon_t$ is publicly observable, and the cancellation result establishes that its conditional variance is regime-invariant: the market maker faces identical informational uncertainty in expansion and contraction. The variance asymmetry therefore originates entirely from loss aversion — the market maker’s behavioural response to its own publicly observable wealth position relative to the Schumpeterian benchmark — not from differential information processing across regimes.*

This is the appropriate baseline for the paper’s identification purpose. With both adverse selection and loss aversion present, the two channels would be observationally entangled in the variance coefficients, making clean separation of λ and g impossible without additional structure. By isolating the loss aversion channel, the paper establishes that it is sufficient for boom-bust asymmetry even in the absence of private information. Robustness Choice 10 confirms that restoring adverse selection scales α and $|\gamma|$ by a common factor $1/(1 + \delta)$, leaving $|\gamma|/\alpha = (\lambda^4 - 1)/\lambda^4$ intact: loss aversion remains the source of asymmetry, adverse selection modulates its magnitude without changing its structure.

Remark 8.2 (News interpretation). *The cancellation result delivers a precise statement about news interpretation in this model. The return innovation $\varepsilon_t^r = \mu_t/2 + \varepsilon_t$ is the publicly observable news signal: its variance is $\bar{\sigma}^2$ regardless of regime, so the same piece of news arrives with identical statistical properties in expansion and contraction. What differs across regimes is not the news itself but its wealth consequence. In expansion, the speculator’s position is $m^{*+} = \lambda\mu_t/(2\lambda_p^0)$; in contraction it is $m^{*-} = \mu_t/(2\lambda\lambda_p^0)$. The ratio is λ^2 : the same signal μ_t generates a position — and therefore a wealth deviation — that is λ^2 times larger in expansion than in contraction. Loss aversion does not distort the interpretation of news; it distorts the response to news, and it does so asymmetrically across the Schumpeterian threshold. This is the model’s formalisation of what news interpretation means in the minimal CPT-like structure.*

8.4 The Threshold-GARCH Coefficients

Theorem 8.3 (Threshold-GARCH for Detrended Wealth Deviations). *Under Assumptions (A1)–(A3), the conditional variance of D_t , $V_t \equiv \text{Var}(D_t \mid \mathcal{F}_t)$, satisfies:*

$$V_t = \omega + \alpha(\varepsilon_t^r)^2 + \gamma(\varepsilon_t^r)^2 \cdot \mathbf{1}[D_{t-1} < 0] + \beta V_{t-1} \quad (10)$$

with exact structural identifications:

$$\alpha = \Phi \lambda^2 > 0 \quad (11)$$

$$\gamma = \Phi \frac{1 - \lambda^4}{\lambda^2} < 0 \quad \text{for all } \lambda > 1 \quad (12)$$

$$\beta = \frac{1}{(1 + g)^2} \in (0, 1) \quad \text{for } g > 0 \quad (13)$$

$$\omega = \frac{2\lambda^2 \sigma_\varepsilon^4 \sigma_\mu^4}{\lambda_p^{02} (4\sigma_\varepsilon^2 + \sigma_\mu^2)^2} \quad (14)$$

where $\Phi \equiv \sigma_\varepsilon^2 \sigma_\mu^2 (16\sigma_\varepsilon^4 + \sigma_\mu^4) / [\lambda_p^{02} (4\sigma_\varepsilon^2 + \sigma_\mu^2)^3]$.

Key structural results: (i) $\gamma = 0$ iff $\lambda = 1$; (ii) $\gamma < 0$ for all $\lambda > 1$ — expansion variance exceeds contraction variance; (iii) $\beta \rightarrow 1$ as $g \rightarrow 0$ (IGARCH boundary); (iv) $|\gamma|/\alpha = (\lambda^4 - 1)/\lambda^4$ depends only on λ ; (v) β depends only on g .

Proof. The exact part follows from Proposition A.6: for each realised regime r , $V_t^r = (B^r)^2 \Sigma[H_0 + H_2(\varepsilon_t^r)^2]$. Setting $\alpha = c_2^+$, $\gamma = c_2^- - c_2^+$, $\omega = c_0^+$, and $\beta = 1/(1 + g)^2$ from the Schumpeterian deflation in (8) gives (10). Signs and the β limit follow from direct differentiation. Full derivation in Proposition A.9. $\square$

Remark 8.4 (Sign of γ and the boom-bust interpretation). *The asymmetry coefficient $\gamma = \Phi(1 - \lambda^4)/\lambda^2$ is strictly negative for all $\lambda > 1$. In the standard GJR-GARCH literature, $\gamma > 0$ reflects the leverage effect: return variance is higher after negative return shocks than after positive ones. The present model has a different object: $V_t = \text{Var}(D_t \mid \mathcal{F}_t)$ is the conditional variance of detrended wealth deviations, not of returns. The correct benchmark is therefore not the leverage effect but the boom-bust asymmetry in wealth dynamics.*

With this benchmark, $\gamma < 0$ is the correct prediction: boom periods (expansion, $D_{t-1} \geq 0$) are more volatile than bust periods (contraction, $D_{t-1} < 0$). This is precisely the pattern documented by Kindleberger (1978) in his historical account of financial manias and by Hamilton (1989) in his Markov-switching characterisation of business cycles: expansions are turbulent and fast, contractions are slow and grinding. The economic mechanism is transparent: in expansion, the loss-averse market maker posts low effective price impact ($\lambda_p^+ = \lambda_p^0/\lambda$), the speculator takes large positions ($m^{+} = \lambda\mu_t/(2\lambda_p^0)$), and the resulting variance of D_t is proportional to $(B^+)^2 = \lambda^2/(4\lambda_p^{02})$. In contraction, the roles are reversed: high price impact, small positions, variance proportional to $(B^-)^2 = 1/(4\lambda^2\lambda_p^{02})$. The ratio is $(B^+)^2/(B^-)^2 = \lambda^4$.*

Theorem 1 delivers exact closed-form coefficients. The following corollary characterises how those coefficients respond to the underlying economic primitives λ , g , and λ_p^0 .

Corollary 8.5 (Comparative statics). *The following partial derivatives hold exactly:*

$$\frac{\partial \beta}{\partial g} = -\frac{2}{(1+g)^3} < 0 \quad (15)$$

$$\frac{\partial |\gamma|}{\partial \lambda} = \frac{2\Phi(\lambda^4 + 1)}{\lambda^3} > 0 \quad (16)$$

$$\frac{\partial |\gamma|}{\partial g} = 0 \quad (17)$$

$$\frac{\partial \alpha}{\partial \lambda} = \frac{2\Phi\lambda}{>} 0 \quad (18)$$

$$\frac{\partial \alpha}{\partial g} = 0 \quad (19)$$

$$\frac{\partial^2 \beta}{\partial g^2} = \frac{6}{(1+g)^4} > 0 \quad (20)$$

Moreover, $\partial \alpha / \partial \lambda_p^0 < 0$ and $\partial |\gamma| / \partial \lambda_p^0 < 0$: both shock-sensitivity coefficients are decreasing in market liquidity.

Proof. All derivatives follow by direct differentiation of the closed-form expressions in Theorem 1, using $\partial \Phi / \partial g = 0$ (since Φ contains only σ_μ , σ_ε , λ_p^0 , and λ). Each sign is immediate from the positivity of all parameters and the facts that $\lambda > 1$ and $g > 0$. All expressions are verified in the companion SymPy notebook. $\square$ $\square$

Four economic implications follow directly from Corollary 8.5.

Separation property. The most striking implication is a clean orthogonality: β depends on g and g alone; $|\gamma|/\alpha = (\lambda^4 - 1)/\lambda^4$ depends on λ and λ alone:

$$\frac{\partial \beta}{\partial \lambda} = 0, \quad \frac{\partial (|\gamma|/\alpha)}{\partial g} = 0.$$

Both are verified in the companion notebook. The persistence parameter β is therefore a sufficient statistic for the Schumpeterian growth rate g , and the asymmetry ratio $|\gamma|/\alpha$ is a sufficient statistic for the loss aversion parameter λ . The two deep parameters are *separately identified* from standard GJR-GARCH output, with no need for auxiliary structural assumptions beyond the model. This clean separation is the paper’s primary identification contribution.

Orthogonal policy channels. Equations (15)–(17) imply that growth-enhancing policies (raising g) reduce volatility persistence β without affecting the asymmetry $|\gamma|$; dealer risk-constraint policies (changing the effective λ via VaR limits or capital requirements) alter $|\gamma|$ without affecting β . The two channels are orthogonal, making identification from observational data cleaner than in models where all GARCH parameters respond to every primitive simultaneously.

Convexity of the persistence effect. The second derivative (20) is positive, meaning the impact of growth on persistence accelerates as g approaches zero. The marginal benefit of growth for volatility stability is largest in stagnant economies — a structural argument for prioritising growth policy in low-growth environments.

Liquidity as a common scaling factor. Both α and $|\gamma|$ are decreasing in λ_p^0 , but their ratio $|\gamma|/\alpha = (\lambda^4 - 1)/\lambda^4$ is independent of λ_p^0 . Market liquidity scales the level of all variance coefficients without affecting their relative magnitude. This means the asymmetry ratio $|\gamma|/\alpha$ can be used to identify λ from cross-market data without requiring knowledge of or controls for market liquidity — a useful robustness property for empirical work.

Table 1 presents numerical comparative statics at the baseline parameterisation.

Table 1: Numerical comparative statics. Base: $\sigma_\mu = 1$, $\sigma_\varepsilon = 0.5$, $\lambda_p^0 = 0.8$, $g = 0.02$.

	α	$ \gamma $	β	$ \gamma /\alpha$
<i>Varying λ ($g = 0.02$):</i>				
$\lambda = 1.0$	0.0977	0.0000	0.9612	0.000
$\lambda = 1.5$	0.2197	0.1763	0.9612	0.803
$\lambda = 2.0$	0.3906	0.3662	0.9612	0.938
$\lambda = 2.25$	0.4944	0.4751	0.9612	0.961
$\lambda = 3.0$	0.8789	0.8681	0.9612	0.988
<i>Varying g ($\lambda = 2.25$):</i>				
$g = 0\%$	0.4944	0.4751	1.0000	0.961
$g = 1\%$	0.4944	0.4751	0.9803	0.961
$g = 2\%$	0.4944	0.4751	0.9612	0.961
$g = 5\%$	0.4944	0.4751	0.9070	0.961
$g = 10\%$	0.4944	0.4751	0.8264	0.961

Note: α and $|\gamma|$ are invariant to g ; β and $|\gamma|/\alpha$ are invariant to λ . The separation property holds exactly.

9 Calibration and the Stationarity Requirement

Proposition A.7 establishes that the Threshold-GARCH process for V_t is covariance-stationary if and only if $\Phi < \Phi_{\max}$, where

$$\Phi_{\max} = \frac{2g\lambda^2(g+2)}{(1+g)^2(\lambda^4+1)}. \quad (21)$$

Substituting the Kyle benchmark $\lambda_p^0 = \sigma_\mu/(2\sigma_u)$, the condition translates into a signal-to-noise requirement on the ratio σ_μ/σ_u of informed signal volatility to noise trader flow volatility. The stationarity condition is therefore a *liquidity condition*: markets with

insufficient informed trading relative to noise trading, combined with high dealer loss aversion and low growth, produce non-stationary volatility in detrended wealth deviations.

Table 2 reports the minimum σ_μ/σ_u required for stationarity across a grid of (λ, g) values, computed at $\sigma_\mu = 1$ and $\sigma_\varepsilon = 0.5$.

Table 2: Minimum σ_μ/σ_u required for covariance-stationarity of V_t . Computed at $\sigma_\mu = 1$, $\sigma_\varepsilon = 0.5$.

λ	$g = 1\%$	$g = 2\%$	$g = 3\%$	$g = 5\%$	$g = 10\%$
1.50	4.13	2.95	2.42	1.90	1.39
2.00	5.19	3.70	3.04	2.39	1.75
2.25	5.78	4.11	3.38	2.66	1.95
2.50	6.38	4.54	3.74	2.94	2.15
3.00	7.60	5.42	4.45	3.50	2.56

At the Tversky-Kahneman parameter value $\lambda = 2.25$ and a standard annual equity growth rate of $g = 2\%$, stationarity requires $\sigma_\mu/\sigma_u > 4.11$. Empirical estimates of the permanent price impact of trades — the empirical counterpart of the model’s signal-to-noise ratio σ_μ/σ_u in the Kyle tradition — consistently exceed this threshold. Hasbrouck (1991) reports estimates of the information content of trades in the range 3–20 for NYSE stocks at the trade-by-trade level, where the information content measures the permanent component of price impact attributable to informed order flow rather than the adverse selection spread component. Large-capitalisation equities, which are the natural empirical counterpart for the model, lie comfortably within the stationary region. The model is therefore empirically operational at the parameter values most relevant for its intended application. The Schumpeterian trend $\bar{r}_t$ plays the role of Hasbrouck (1991)’s quote midpoint in this framework: both are the publicly observable efficient benchmark from which transaction-level deviations are measured and to which they mean-revert. The mean-reversion speed $1 - \beta = 1 - 1/(1 + g)^2$ in the deviation recursion corresponds directly to Hasbrouck’s transient price impact component.

Remark 9.1 (VAR structure). *The deviation recursion $D_t = D_{t-1}/(1+g) + A^r\sigma_\mu^2 + B^r\mu_t\varepsilon_t$ is structurally an AR(1) with autoregressive coefficient $1/(1+g)$ and regime-dependent innovation. This is the structural counterpart of the bivariate VAR that Hasbrouck (1991) estimates empirically: the permanent component of any shock to D_t is zero under stationarity ($g > 0$) because all wealth deviations eventually revert to the Schumpeterian benchmark, while the transient component decays geometrically at rate $1/(1+g)$ per period. At the IGARCH boundary ($g \rightarrow 0$) the autoregressive coefficient approaches one and shocks become permanent, giving a unit-root VAR. The Threshold-GARCH variance recursion for V_t then overlays a regime-dependent second-moment structure on top of this AR foundation — an extension beyond Hasbrouck’s constant-variance framework. The natural empirical strategy is therefore a regime-switching extension of the Hasbrouck (1991)*

VAR with threshold at the Schumpeterian trend crossing $D_{t-1} = 0$, estimating β from the autoregressive coefficient and the variance ratio λ^4 from the regime-dependent innovation variances.

Three further properties of (21) are worth noting. First, $\Phi_{\max} \rightarrow 0$ as $g \rightarrow 0$: for any fixed $\Phi > 0$, the stationarity condition eventually fails as growth vanishes. This is precisely the IGARCH boundary of Proposition A.8 — non-stationary volatility in detrended wealth deviations is the structural signature of a stagnant Schumpeterian economy. Second, $\Phi_{\max}$ is decreasing in λ : more loss-averse dealers impose a higher liquidity requirement for stationarity, reflecting the amplified regime asymmetry encoded in the λ^4 ratio of Proposition A.6. Third, at $\lambda = 1$ (loss-neutral dealers), $\Phi_{\max} = g(g+2)/(1+g)^2$, which is positive for all $g > 0$ and easily achievable, confirming that the stationarity constraint is entirely a consequence of dealer loss aversion rather than a feature of the Schumpeterian growth structure alone.

The persistence parameter $\beta = 1/(1+g)^2$ is plotted against g in the text and takes the values 0.980, 0.961, 0.943, 0.907, and 0.826 at $g \in \{1\%, 2\%, 3\%, 5\%, 10\%\}$ respectively. These are consistent with empirical GARCH estimates for equity markets, which typically find $\beta \in (0.85, 0.98)$, and provide a structural interpretation: markets with higher long-run growth rates should exhibit lower volatility persistence, a prediction that is testable in cross-country panel data.

10 Robustness: Verifying the Necessary Structural Conditions

The structural results rest on two necessary conditions: the shared Schumpeterian reference point (which, together with zero-sum dynamics and position closing, is necessary for regime separation) and loss aversion $\lambda > 1$ in the value function (which is necessary and sufficient for the variance asymmetry). This section examines how the boom-bust asymmetry and the persistence identification are affected when each of the eleven design choices is relaxed, verifying that these two conditions are necessary rather than merely convenient. The key finding is that $\partial|\gamma|/\partial\lambda > 0$ — more dealer loss aversion always produces more boom-bust asymmetry — survives all eleven relaxations. The $\partial\beta/\partial g < 0$ direction fails only when the two necessary conditions are directly removed.

Effect of each design choice on the structural results

Choice 1 — Linear price impact. Nonlinear price impact breaks the exact cancellation result: the product $\lambda_p^{\text{eff}} \cdot m_t^*$ is no longer $\mu_t/2$ in both regimes. The variance recursion becomes a Power ARCH (PARCH) specification. The boom-bust variance asymmetry

survives because the regime-dependent bound structure is preserved; both comparative static directions are maintained.

Choice 2 — Quadratic cost (price impact). Without price impact ($\lambda_p^0 \rightarrow 0$), the speculator's problem has no finite interior solution and the β channel collapses. Result: Threshold-ARCH(1). The Schumpeterian mean-reversion channel — the structural source of $\beta = f(g)$ — is disrupted. The $\partial|\gamma|/\partial\lambda > 0$ direction is preserved. This choice is *a necessary condition for the persistence identification*.

Choice 3 — Position closing. Carry-over positions introduce additional lags, extending the recursion to Threshold-GARCH $(n, 1)$. Both comparative static directions are preserved in each lag.

Choice 4 — Zero-sum condition. Spread income $s > 0$ breaks zero-sum and admits a third regime in which both agents can simultaneously be on the same side of the kink. Result: three-regime asymmetric variance recursion. The $\partial\beta/\partial g < 0$ direction is preserved; the loss aversion effect is enriched with three separate λ -dependent asymmetry terms.

Choice 5 — Regime structure. The regime structure is inherited from Choices 3, 4, and 6. When the shared reference point is intact, regime separation holds and the variance asymmetry is preserved regardless of the number of lags or the presence of spread income.

Choice 6 — Shared reference point. Replacing the shared Schumpeterian trend with own-history reference points eliminates the Schumpeterian deflation from the deviation recursion. The persistence parameter becomes $\beta = \rho^2$ where ρ is the own-history mean-reversion coefficient, severing the structural link between β and g . This choice is *a necessary condition for the persistence identification*: removing it eliminates the paper's second main result.

Choice 7 — Identical λ . Heterogeneous loss aversion $\lambda^{MM} \neq \lambda^S$: the variance asymmetry is driven by λ^{MM} alone, since λ^S cancels in the speculator's first-order condition (Remark 2.2). Both comparative static directions are preserved.

Choice 8 — Constant g . Time-varying g_t delivers $\beta_t = 1/(1 + g_t)^2$ period by period, yielding a Markov-Switching Threshold-GARCH with growth cycles as the state variable. The separation property holds period by period: β_t depends only on g_t and the asymmetry ratio depends only on λ .

Choice 9 — MM zero-profit benchmark. A monopolist MM yields an identical λ_p^0 in the linear Kyle framework. Both comparative static directions are fully preserved.

Choice 10 — Public signal. Under private information, adverse selection scales both α and $|\gamma|$ uniformly by $1/(1 + \delta)$ where δ is the adverse selection premium. The ratio $|\gamma|/\alpha = (\lambda^4 - 1)/\lambda^4$ is preserved exactly, as is the separation property.

Choice 11 — Stationarity ($g > 0$). As $g \rightarrow 0$, $\beta \rightarrow 1$ (IGARCH boundary, Proposition A.8). The $\partial\beta/\partial g$ derivative diverges at $g = 0$, reflecting that the persistence effect is strongest precisely in stagnant economies. The $\partial|\gamma|/\partial\lambda > 0$ direction is preserved, and $|\gamma|$ is independent of g (verified symbolically).

Table 3 summarises the robustness of both comparative static directions across all eleven choices.

Table 3: Robustness of design choices: effect on structural identifications.

Choice Relaxation		Variance re- cursion	$\partial\beta/\partial g < 0?$	$\partial \gamma /\partial\lambda > 0?$
1	Nonlinear price impact	PARCH	Preserved	Preserved
2	No quadratic cost	Threshold-ARCH	Disrupted	Preserved
3	Carry-over positions	T-GARCH($n, 1$)	Preserved	Preserved
4	Spread income ($s > 0$)	3-regime GARCH	Preserved	Enriched
5	Regime structure	Inherited	Inherited	Inherited
6	Own-history ref point	$\beta = \rho^2$	Disrupted	Preserved
7	Het. λ	T-GARCH	Preserved	Preserved
8	Time-varying g_t	MS-T-GARCH	Preserved (p-b-p)	Preserved
9	Monopolist MM	T-GARCH	Preserved	Preserved
10	Private signal	Scaled T-GARCH	Preserved	Preserved
11	$g \rightarrow 0$	T-IGARCH	Boundary	Preserved

Note: T-GARCH = Threshold-GARCH; MS = Markov-Switching; p-b-p = period by period. Choices 2 and 6 remove the necessary conditions for the $\beta = f(g)$ identification. The $\partial|\gamma|/\partial\lambda > 0$ direction — more dealer loss aversion produces more boom-bust asymmetry — survives all eleven relaxations, confirming it is the most robust result in the paper. Both conditions are necessary rather than merely convenient: removing either eliminates one of the two main structural identifications.

11 Conclusion

11.1 Summary

This paper identifies the minimal behavioural structure sufficient to derive the boom-bust asymmetry in financial wealth dynamics as an equilibrium theorem. We retain the single empirically robust element of prospect theory — the kink in the value function at a reference point — and embed it in a Stackelberg market with a Schumpeterian background growth trend as the shared reference point.

Two structural features are jointly necessary conditions. First, the combination of zero-sum dynamics, the shared reference point, and within-period position closing guarantees that the market maker and speculator always sit on opposite sides of the kink simultaneously — a derived theorem, not an assumption. This generates the boom-bust

variance asymmetry: expansions are λ^4 times more volatile in detrended wealth deviations than contractions, where λ is the dealer's loss aversion parameter. Second, the Schumpeterian trend provides the mean-reversion that pins volatility persistence to the economy's long-run growth rate g through the exact structural mapping $\beta = 1/(1 + g)^2$.

The separation property is the paper's primary identification result: β depends on g alone (independent of λ) and $|\gamma|/\alpha = (\lambda^4 - 1)/\lambda^4$ depends on λ alone (independent of g). The two structural primitives — growth and dealer loss aversion — are separately identified from the shape and persistence of observed wealth volatility dynamics, without auxiliary assumptions.

The paper began with the goal of finding micro-foundations of GARCH behaviour through a minimal CPT-like structure. That goal is achieved, but with a precise qualification: the GARCH-type recursion is derived for the conditional variance of detrended wealth deviations rather than directly for equity return variance, which is the conventional object of empirical GARCH estimation. This is not a failure of the approach but a clarification of what the minimal structure delivers. The cancellation result — that return variance is regime-invariant — shows that the same piece of news arrives identically in both regimes; what differs is the wealth consequence of that news, which is λ^2 times larger in expansion than in contraction. Boom-bust asymmetry in financial wealth is therefore the correct and precise object of the micro-foundation, and the paper delivers it as an equilibrium theorem rather than a calibration choice. The connection to return variance is a direction for future work.

11.2 The IGARCH boundary and its interpretation

As $g \rightarrow 0$, $\beta \rightarrow 1$ and the conditional variance of detrended wealth deviations follows a Threshold-IGARCH process. This provides the first preference-based micro-foundation for integrated volatility processes. Near-unit-root volatility persistence — documented in low-growth episodes such as post-2008 Japan and post-crisis Europe — is not an arbitrary statistical restriction but the volatility regime of a stagnant Schumpeterian economy.

Importantly, the asymmetry coefficient γ is independent of g : $\partial\gamma/\partial g = 0$. A stagnant economy therefore exhibits the same boom-bust asymmetry as a growing one, but with permanently persistent volatility shocks. The two features — asymmetry and persistence — are independently determined by independent structural primitives.

11.3 Why loss aversion must reside in the market maker

The assignment of loss aversion to the market maker is not a modelling choice but a structural necessity established in Remark 2.2: loss aversion in the speculator's objective cancels in the first-order condition and generates no asymmetry. The mechanism

requires loss aversion to enter through a binding constraint on the price-setter, which then propagates to the follower’s optimal quantity choice.

This is independently confirmed at the desk level by Li et al. (2025), who document that US Treasury dealers reduce position changes and require higher compensation as they approach their internal VaR limits. The model’s contraction regime — high price impact, small speculator positions, compressed wealth volatility — matches this documented behaviour precisely.

11.4 Testable predictions and the identification programme

Prediction 0 (empirical strategy). The structural recursion can be taken to data using a regime-switching extension of the Hasbrouck (1991) VAR framework. Estimate a bivariate system in detrended wealth or dealer balance sheet deviations and regime indicator, with threshold at the Schumpeterian trend crossing $D_{t-1} = 0$. The autoregressive coefficient in the deviation equation identifies $\beta = 1/(1+g)^2$ and the ratio of regime-specific innovation variances identifies λ^4 . The cross-sectional restriction — β co-varies with long-run TFP growth and λ^4 co-varies with experimental loss aversion estimates across countries, with zero cross-effects — is the primary test of the separation property.

The separation property generates a falsifiable cross-sectional restriction invisible from single-market time-series estimation. Across markets $i = 1, \dots, N$, the model predicts:

$$\hat{\beta}_i = \frac{1}{(1+g_i)^2} + u_i, \quad \frac{|\hat{\gamma}_i|}{\hat{\alpha}_i} = \frac{\lambda_i^4 - 1}{\lambda_i^4} + v_i \quad (22)$$

where the two equations share no common primitive: g_i enters only the first and λ_i only the second. The natural empirical test uses Penn World Tables long-run TFP growth for g_i , cross-cultural experimental estimates of loss aversion for λ_i (Wang et al., 2017, provide 53-country evidence), and Threshold-GARCH estimation for $\hat{\beta}_i$, $\hat{\alpha}_i$, $|\hat{\gamma}_i|$ on detrended wealth or dealer balance sheet data.

The policy implication of the orthogonality is direct. Growth-enhancing structural reforms reduce wealth volatility persistence β without affecting asymmetry $|\gamma|/\alpha$. Macroprudential policies that constrain dealer risk-taking — tighter VaR limits, higher capital requirements — reduce boom-bust asymmetry without affecting persistence. The two policy instruments operate through different structural channels and are therefore non-redundant: a result that cannot be derived from reduced-form estimation.

11.5 Policy implications

The orthogonality of the two structural identifications generates a precise policy implication. Because $\beta = f(g)$ only and $|\gamma|/\alpha = f(\lambda)$ only, growth-enhancing structural reforms and macroprudential policies that constrain dealer risk-taking are *non-redundant*

instruments: they target different features of wealth volatility dynamics through different structural channels.

This connects to the practitioner debate on procyclicality in financial markets. Adrian and Shin (2010, 2014) document that dealer balance sheets expand rapidly above trend and contract slowly below it — precisely the λ^4 variance asymmetry the model quantifies. The model says that the amplitude of this procyclicality is governed by dealer loss aversion λ , not by the growth rate. A macroprudential policy that reduces effective λ — through tighter VaR limits, higher capital requirements, or countercyclical capital buffers — reduces $|\gamma|/\alpha$ without affecting β . Conversely, a growth-enhancing reform that raises g reduces β — shortening the half-life of volatility shocks in wealth dynamics — without affecting the boom-bust asymmetry.

The structural model therefore provides a micro-founded rationale for why both instruments are simultaneously necessary: macroprudential policy addresses the cross-sectional asymmetry (λ channel) while growth policy addresses persistence (g channel). Neither instrument substitutes for the other, and the model quantifies the magnitude of each effect through the exact structural mappings of Theorem 1 and Corollary 8.5. This is a structural claim about the architecture of financial volatility that reduced-form estimation cannot deliver.

11.6 Limitations and future directions

Three limitations should be stated clearly.

First, the persistence identification is a leading-order approximation. The exact expression for β involves the normal density evaluated at the contraction threshold, and the approximation error is bounded at under 6% for moderate deviations from the stationary point (Proposition A.5). The structural mapping $\beta = 1/(1+g)^2$ is the leading-order term; all qualitative results and the separation property hold exactly.

Second, the paper derives a structural recursion for the conditional variance of *detrended wealth deviations* $V_t = \text{Var}(D_t \mid \mathcal{F}_t)$, not for the conditional variance of equity returns. The cancellation result (Section 8.3) establishes that the one-period return variance is $\bar{\sigma}^2$ in both regimes — regime-invariant — so the structural asymmetry operates through wealth dynamics rather than return dynamics. The connection between wealth deviation variance and return variance, while structurally motivated, requires a bridge argument that is left for future work. The object of the paper — boom-bust asymmetry in financial wealth — is the direct empirical counterpart of the documented patterns in dealer balance sheet dynamics (Adrian and Shin, 2010, 2014; Brunnermeier and Pedersen, 2009) and in post-crisis wealth recovery paths (Reinhart and Rogoff, 2009).

Third, the reference point construction — a common Schumpeterian trend at a constant growth rate — abstracts from heterogeneous benchmarks, time-varying growth, and

catching-up dynamics. Time-varying g_t following a Schumpeterian wave process is the most important extension: it delivers Markov-Switching Threshold-GARCH with growth cycles as the state variable, connecting the framework to Hamilton-style regime switching in growth rates. Heterogeneous loss aversion $\lambda^{MM} \neq \lambda^S$ and carry-over positions (extending to Threshold-GARCH $(n, 1)$) are further natural extensions.

A Technical Appendix

This appendix collects all proofs omitted from the main text. The structure mirrors the main text: Lemmas A.1–A.4 establish the conditional distribution machinery; Propositions A.1–A.5 prove the main results; Proposition A.5 states the linearisation approximation with explicit error bounds. All symbolic results have been verified in SymPy (see the companion verification notebook).

Standing assumptions throughout the appendix.

- (A1) *Joint Gaussianity.* $\mu_t \sim \mathcal{N}(0, \sigma_\mu^2)$ and $\varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ are mutually independent. The return innovation is $\varepsilon_t^r = \frac{1}{2}\mu_t + \varepsilon_t$.
- (A2) *Deviation recursion.* $D_t = D_{t-1}/(1+g) + A^r \sigma_\mu^2 + B^r \mu_t \varepsilon_t$, where $r \in \{+, -\}$ denotes the regime determined by $\text{sign}(D_{t-1})$, and A^r, B^r are the regime-specific coefficients of equations (38)–(39).
- (A3) *Isserlis.* For any zero-mean bivariate Gaussian (X, Y) : $\mathbb{E}[X^2 Y^2] = \text{Var}(X) \text{Var}(Y) + 2 \text{Cov}(X, Y)^2$.

A.1 Conditional distribution lemmas

Lemma A.1 (Conditional distribution of (μ_t, ε_t) given ε_t^r). *Under Assumption (A1), the conditional distribution is bivariate Gaussian with moments:*

$$\mathbb{E}[\mu_t \mid \varepsilon_t^r] = \rho_\mu \varepsilon_t^r, \quad \mathbb{E}[\varepsilon_t \mid \varepsilon_t^r] = \rho_\varepsilon \varepsilon_t^r \quad (23)$$

$$\text{Var}(\mu_t \mid \varepsilon_t^r) = \frac{4\sigma_\varepsilon^2 \sigma_\mu^2}{4\sigma_\varepsilon^2 + \sigma_\mu^2}, \quad \text{Var}(\varepsilon_t \mid \varepsilon_t^r) = \frac{\sigma_\varepsilon^2 \sigma_\mu^2}{4\sigma_\varepsilon^2 + \sigma_\mu^2} \quad (24)$$

$$\text{Cov}(\mu_t, \varepsilon_t \mid \varepsilon_t^r) = \frac{-2\sigma_\varepsilon^2 \sigma_\mu^2}{4\sigma_\varepsilon^2 + \sigma_\mu^2} \quad (25)$$

where $\rho_\mu \equiv \sigma_\mu^2/(2\bar{\sigma}^2)$, $\rho_\varepsilon \equiv \sigma_\varepsilon^2/\bar{\sigma}^2$, and $\bar{\sigma}^2 \equiv \sigma_\mu^2/4 + \sigma_\varepsilon^2$.

Proof. By standard Gaussian conditioning. Since (μ_t, ε_t) are jointly Gaussian and independent, and $\varepsilon_t^r = \frac{1}{2}\mu_t + \varepsilon_t$ is a linear combination, $(\mu_t, \varepsilon_t, \varepsilon_t^r)$ is multivariate Gaussian. The cross-covariances are $\text{Cov}(\mu_t, \varepsilon_t^r) = \sigma_\mu^2/2$ and $\text{Cov}(\varepsilon_t, \varepsilon_t^r) = \sigma_\varepsilon^2$. Dividing

by $\text{Var}(\varepsilon_t^r) = \bar{\sigma}^2$ gives (23). The conditional variances and covariance follow from the Schur complement formula applied to the joint covariance matrix. *Consistency check:* $\frac{1}{2}\rho_\mu + \rho_\varepsilon = \sigma_\mu^2/(4\bar{\sigma}^2) + \sigma_\varepsilon^2/\bar{\sigma}^2 = \bar{\sigma}^2/\bar{\sigma}^2 = 1$, confirming $\mathbb{E}[\varepsilon_t^r | \varepsilon_t^r] = \varepsilon_t^r$. $\square$ $\square$

Lemma A.2 (Conditional expectation of $\mu_t \varepsilon_t$). *Under Assumption (A1):*

$$\mathbb{E}[\mu_t \varepsilon_t | \varepsilon_t^r] = \rho_\mu [\rho_\varepsilon (\varepsilon_t^r)^2 - \sigma_\varepsilon^2] \quad (26)$$

This expression is even in ε_t^r : it depends only on $(\varepsilon_t^r)^2$, not on the sign of ε_t^r .

Proof. Decompose $\mu_t \varepsilon_t = \text{Cov}(\mu_t, \varepsilon_t | \varepsilon_t^r) + \mathbb{E}[\mu_t | \varepsilon_t^r] \cdot \mathbb{E}[\varepsilon_t | \varepsilon_t^r]$. Substituting (25) and (23):

$$\mathbb{E}[\mu_t \varepsilon_t | \varepsilon_t^r] = -\frac{2\sigma_\varepsilon^2 \sigma_\mu^2}{4\sigma_\varepsilon^2 + \sigma_\mu^2} + \rho_\mu \varepsilon_t^r \cdot \rho_\varepsilon \varepsilon_t^r = -\rho_\mu \sigma_\varepsilon^2 + \rho_\mu \rho_\varepsilon (\varepsilon_t^r)^2 = \rho_\mu [\rho_\varepsilon (\varepsilon_t^r)^2 - \sigma_\varepsilon^2].$$

Evenness follows immediately since the right-hand side contains only $(\varepsilon_t^r)^2$. $\square$ $\square$

Lemma A.3 ($\text{Var}(\mu_t \varepsilon_t | \varepsilon_t^r)$ via Isserlis). *Under Assumptions (A1) and (A3):*

$$\text{Var}(\mu_t \varepsilon_t | \varepsilon_t^r) = \Sigma [H_0 + H_2 (\varepsilon_t^r)^2] \quad (27)$$

where $\Sigma \equiv \sigma_\varepsilon^2 \sigma_\mu^2 / (4\sigma_\varepsilon^2 + \sigma_\mu^2)^2$, $H_0 \equiv 8\sigma_\varepsilon^2 \sigma_\mu^2$, and $H_2 \equiv 4(16\sigma_\varepsilon^4 + \sigma_\mu^4) / (4\sigma_\varepsilon^2 + \sigma_\mu^2)$. In particular $\text{Var}(\mu_t \varepsilon_t | \varepsilon_t^r) > 0$ and both $H_0 > 0$ and $H_2 > 0$.

Proof. Let $u \equiv \mu_t - \mathbb{E}[\mu_t | \varepsilon_t^r]$ and $v \equiv \varepsilon_t - \mathbb{E}[\varepsilon_t | \varepsilon_t^r]$ be the conditional residuals, with $(u, v) \sim \mathcal{N}(0, V)$ where V is the 2×2 conditional covariance matrix of Lemma A.1. Denote conditional means $m_\mu = \rho_\mu \varepsilon_t^r$ and $m_\varepsilon = \rho_\varepsilon \varepsilon_t^r$. Then $\mu_t \varepsilon_t = (u + m_\mu)(v + m_\varepsilon)$. Expanding:

$$\mathbb{E}[(\mu_t \varepsilon_t)^2 | \varepsilon_t^r] = \mathbb{E}[u^2 v^2] + m_\varepsilon^2 \text{Var}(u) + m_\mu^2 \text{Var}(v) + (m_\mu m_\varepsilon)^2 + 2m_\mu m_\varepsilon \text{Cov}(u, v),$$

where all odd moments of (u, v) vanish by Gaussianity. Applying Assumption (A3): $\mathbb{E}[u^2 v^2] = \text{Var}(u) \text{Var}(v) + 2 \text{Cov}(u, v)^2$. Subtracting $(\mathbb{E}[\mu_t \varepsilon_t | \varepsilon_t^r])^2$ and collecting terms in $(\varepsilon_t^r)^0$ and $(\varepsilon_t^r)^2$ yields (27). Positivity of H_0 and H_2 is immediate from the positivity of all parameters. $\square$ $\square$

Lemma A.4 (Slope cancellation). *The partial derivative of the standardised threshold $z_t = -\mu_{D_t}^r / \psi_t^r$ with respect to $(\varepsilon_t^r)^2$ is regime-invariant:*

$$\frac{\partial z_t}{\partial (\varepsilon_t^r)^2} = -\frac{\rho_\mu \rho_\varepsilon}{\sqrt{\Sigma H_0}} \quad \text{for both } r \in \{+, -\}.$$

Proof. From Lemma A.2, $\delta^r \equiv \partial \mu_{D_t}^r / \partial (\varepsilon_t^r)^2 = B^r \rho_\mu \rho_\varepsilon$. The conditional standard deviation of D_t at $\varepsilon_t^r = 0$ is $\psi_0^r = B^r \sqrt{\Sigma H_0}$. Therefore $\partial z_t / \partial (\varepsilon_t^r)^2 = -\delta^r / \psi_0^r = -B^r \rho_\mu \rho_\varepsilon / (B^r \sqrt{\Sigma H_0}) = -\rho_\mu \rho_\varepsilon / \sqrt{\Sigma H_0}$, where B^r cancels exactly in both regimes. $\square$ $\square$

A.2 Main propositions

Proposition A.5 (Asymmetric drift). *Define the regime-specific stationary points $D_r^* = A^r \sigma_\mu^2(1+g)/g$. Then $D_+^*/D_-^* = \lambda^2$ and both are strictly positive. The process $\{D_t\}$ satisfies:*

- (i) *In contraction ($D_{t-1} < 0$): $\mathbb{E}[D_t | D_{t-1}] > D_{t-1}$ unconditionally (submartingale).*
- (ii) *In expansion ($D_{t-1} \geq 0$): $\mathbb{E}[D_t | D_{t-1}] \geq D_{t-1}$ as $D_{t-1} \leq D_+^*$.*
- (iii) *$D_+^* > D_-^* > 0$: the economy has a positive bias.*
- (iv) *The deflation factor $1/(1+g)$ is identical in both regimes; asymmetry enters only through the drift $A^r \sigma_\mu^2$.*

Proof. By Assumption (A1), $\mathbb{E}[\mu_t \varepsilon_t] = 0$ (independence and zero means), so $\mathbb{E}[D_t | D_{t-1}] = D_{t-1}/(1+g) + A^r \sigma_\mu^2$. The fixed point $\mathbb{E}[D_t | D_{t-1}] = D_{t-1}$ gives $D_r^* = A^r \sigma_\mu^2(1+g)/g > 0$ since $A^r, \sigma_\mu^2, g > 0$. The ratio $D_+^*/D_-^* = A^+/A^- = \lambda^2$ follows from equations (38)–(39).

Part (i). For $D_{t-1} < 0$: $\mathbb{E}[D_t | D_{t-1}] - D_{t-1} = -D_{t-1}g/(1+g) + A^- \sigma_\mu^2 > 0$, since both terms are strictly positive.

Part (ii). For $D_{t-1} \geq 0$: the same expression has sign equal to $\text{sign}(D_+^* - D_{t-1})$.

Parts (iii) and (iv). Both follow directly from the definitions and $A^+/A^- = \lambda^2 > 1$. □

Proposition A.6 (Variance asymmetry). *Under Assumptions (A1)–(A3), the conditional variance of D_t given $(\varepsilon_t^r, D_{t-1})$ satisfies:*

$$\text{Var}(D_t | \varepsilon_t^r, D_{t-1}) = (B^r)^2 \Sigma[H_0 + H_2(\varepsilon_t^r)^2] \quad (28)$$

and the ratio across regimes is:

$$\frac{\text{Var}(D_t | \varepsilon_t^r, D_{t-1}, D_{t-1} \geq 0)}{\text{Var}(D_t | \varepsilon_t^r, D_{t-1}, D_{t-1} < 0)} = \frac{(B^+)^2}{(B^-)^2} = \lambda^4.$$

Proof. From Assumption (A2), $D_t - D_{t-1}/(1+g) - A^r \sigma_\mu^2 = B^r \mu_t \varepsilon_t$. Since D_{t-1} and ε_t^r are $\mathcal{F}_{t-1}$ -measurable or determined by the current period's shock distribution, and B^r is $\mathcal{F}_{t-1}$ -measurable:

$$\text{Var}(D_t | \varepsilon_t^r, D_{t-1}) = (B^r)^2 \text{Var}(\mu_t \varepsilon_t | \varepsilon_t^r).$$

Substituting Lemma A.3 gives (28). The ratio $(B^+)^2/(B^-)^2 = [\lambda/(2\lambda_p^0)]^2/[1/(2\lambda_p^0)]^2 = \lambda^4$ follows directly. □

Proposition A.7 (Stationarity condition). *The Threshold-GARCH process for V_t is covariance-stationary if and only if:*

$$\Phi < \Phi_{\max} \equiv \frac{2g\lambda^2(g+2)}{(1+g)^2(\lambda^4+1)} \quad (29)$$

Three corollaries: (a) $\Phi_{\max} \rightarrow 0$ as $g \rightarrow 0$, recovering the IGARCH boundary; (b) at $\lambda = 1$, $\Phi_{\max} = g(g+2)/(1+g)^2 > 0$; (c) $\Phi_{\max}$ is strictly decreasing in λ .

Proof. The standard covariance-stationarity condition for a Threshold-GARCH process is $\alpha + \gamma\pi^* + \beta < 1$, where π^* is the ergodic probability of contraction. At the symmetric stationary point $\pi^* = 1/2$, this becomes $\alpha + \gamma/2 + \beta < 1$. From Theorem 1: $\alpha + \gamma/2 = \Phi(\lambda^4 + 1)/(2\lambda^2)$. Substituting $\beta = 1/(1+g)^2$ and solving for Φ gives (29). Corollary (a): substitute $g = 0$. Corollary (b): substitute $\lambda = 1$ and simplify. Corollary (c): $\partial\Phi_{\max}/\partial\lambda = 2g(g+2)(1+g)^{-2} \cdot \partial[\lambda^2/(\lambda^4+1)]/\partial\lambda$; the bracketed expression has derivative $\lambda(2-2\lambda^4)/(\lambda^4+1)^2 < 0$ for $\lambda > 1$. $\square$ $\square$

Proposition A.8 (IGARCH boundary). *As $g \rightarrow 0$, $\beta = 1/(1+g)^2 \rightarrow 1$ and the process converges to a Threshold-IGARCH with $\alpha + \gamma/2 + \beta = 1$. The asymmetry coefficient γ satisfies $\partial\gamma/\partial g = 0$: it is independent of the growth rate. A stagnant economy therefore exhibits the same leverage asymmetry as a growing one, but with permanently persistent variance shocks.*

Proof. $\beta = 1/(1+g)^2 \rightarrow 1$ as $g \rightarrow 0$ is immediate. That $\partial\gamma/\partial g = 0$ follows because $\gamma = \Phi(1-\lambda^4)/\lambda^2$ and Φ does not depend on g (it contains only σ_μ , σ_ε , λ_p^0 , and λ). $\square$ $\square$

A.3 Linearisation and error bounds

Proposition A.9 (Linearisation approximation and error bound). **Exact part.** *For each realised regime $r \in \{+, -\}$, the conditional variance of D_t is:*

$$V_t^r \equiv \text{Var}(D_t \mid \varepsilon_t^r, D_{t-1}, \text{regime} = r) = (B^r)^2 \Sigma [H_0 + H_2(\varepsilon_t^r)^2]. \quad (30)$$

This holds exactly under Assumptions (A1)–(A3).

Approximate part. *The expected conditional variance $\mathbb{E}[V_t \mid \mathcal{F}_{t-1}]$ equals $\pi_{t-1}V_t^- + (1 - \pi_{t-1})V_t^+$, where $\pi_{t-1} = \Phi(z_{t-1})$ and z_{t-1} is the standardised threshold. Define $\bar{V} \equiv \Sigma(H_0 + H_2\bar{\sigma}^2) = \sigma_\varepsilon^2\sigma_\mu^2$ and expand $\Phi(z_{t-1})$ to first order around $(D_{t-2} = D_r^*, \varepsilon_{t-1}^r = 0)$:*

$$\pi_{t-1} \approx \pi^* + \phi(z^*) \left[-\frac{D_{t-2} - D_r^*}{(1+g)\psi_0^r} - \frac{\rho_\mu\rho_\varepsilon}{\sqrt{\Sigma H_0}}(\varepsilon_{t-1}^r)^2 \right] \quad (31)$$

where $\psi_0^r = B^r\sqrt{\Sigma H_0}$. Substituting (31) into $\mathbb{E}[V_t \mid \mathcal{F}_{t-1}]$ and identifying $V_{t-1} \propto \pi_{t-2}$.

$[(B^-)^2 - (B^+)^2]\bar{V}$ yields a recursion of the form:

$$\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] \approx \omega + \alpha(\varepsilon_{t-1}^r)^2 + \gamma(\varepsilon_{t-1}^r)^2 \mathbf{1}[D_{t-2} < 0] + \beta V_{t-1} \quad (32)$$

with $\beta = 1/(1+g)^2$ arising from the $D_{t-2}/(1+g)$ term in z_{t-1} , which propagates through two lags to V_{t-1} .

Error bound. The error in approximating π_{t-1} by (31) satisfies, by the Taylor remainder theorem:

$$|\pi_{t-1} - \pi_{t-1}^{\text{approx}}| \leq \frac{\phi(0)}{2\sqrt{e}} (z_{t-1} - z^*)^2.$$

For deviations $|D_{t-2} - D_r^*| \leq \delta \psi_0^r(1+g)$, the implied error in $\mathbb{E}[V_t \mid \mathcal{F}_{t-1}]$ is:

$$|\mathbb{E}[V_t \mid \mathcal{F}_{t-1}] - \mathbb{E}[V_t \mid \mathcal{F}_{t-1}]^{\text{approx}}| \leq \bar{V} |(B^-)^2 - (B^+)^2| \frac{\phi(0)}{2\sqrt{e}} \delta^2.$$

At $\lambda = 2.25$, $\lambda_p^0 = 0.8$, $\sigma_\mu = 1$, $\sigma_\varepsilon = 0.5$, and $\delta = 0.5$, the relative error is 5.8%. The approximation is accurate when the economy is within $O(\psi_0^r)$ of the stationary point — the empirically relevant regime for moderate deviations from trend.

Proof. Exact part. Follows directly from Proposition A.6 (Lemma A.3 and Assumption (A2)).

Error bound. The function $\Phi : \mathbb{R} \rightarrow (0, 1)$ is twice continuously differentiable with $\Phi'' = -z\phi(z)$. By Taylor's theorem with remainder: $|\Phi(z) - \Phi(z^*) - \phi(z^*)(z - z^*)| \leq \sup_\xi |\Phi''(\xi)| \cdot (z - z^*)^2/2$. Since $|\Phi''(z)| = |z|\phi(z) \leq \phi(0)/\sqrt{e}$ (attained at $z = \pm 1$), the bound follows. The remainder maps to the V_t recursion by multiplying through by $|(B^-)^2 - (B^+)^2| \cdot \bar{V}$. $\square$ $\square$

A.4 Summary table

Table 4 records every result verified symbolically in the companion SymPy notebook (`threshold_garch_verification_v3.ipynb`).

Statements and Declarations

Competing interests. The author declares no competing interests.

Funding. No external funding was received for this research.

Data availability. No datasets were generated or analysed during the current study. The SymPy verification notebook supporting the symbolic results is available from the author on reasonable request.

Table 4: SymPy verification record.

Block	Result	Status
1	Lemma A.1: conditional distribution $(\mu_t, \varepsilon_t) \mid \varepsilon_t^r$	PASS
2	Lemma A.2: $\mathbb{E}[\mu_t \varepsilon_t \mid \varepsilon_t^r]$ even in ε_t^r	PASS
3	Lemma A.3: $\text{Var}(\mu_t \varepsilon_t \mid \varepsilon_t^r)$ via Isserlis	PASS
4	Proposition 7a: deviation projection κ^r, δ^r	PASS
5	Proposition 7: all λ^2 ratios $(\kappa, \delta, \lambda_p, m^*)$	PASS
6	Lemma A.4: slope cancellation	PASS
7	Proposition A.6: $\text{Var}(D_t)$ ratio = λ^4	PASS
8	Theorem 1: exact $\alpha, \gamma, \beta, \omega$	PASS
9	Proposition A.5: asymmetric drift, $D_+^*/D_-^* = \lambda^2$	PASS
10	Proposition A.7: $\Phi < \Phi_{\max}$	PASS
11	Proposition A.8: IGARCH boundary, $\partial\gamma/\partial g = 0$	PASS
12	Calibration: minimum σ_μ/σ_u at Tversky-Kahneman values	PASS

51 individual assertions, 0 errors. SymPy version 1.14.0.

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